`RichnessSelection`: counting the eligible galaxies behind the projection integrals

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Contents

1 The physical question: which galaxies contaminate $\lambda^{\rm ob}$?

redMaPPer estimates a cluster's richness by counting member galaxies inside a disk of angular radius θ_λ and selecting them with an effective photo-z window around the cluster, $w(z, z^{\rm ob})$: the probability that a galaxy at redshift z passes the photo-z cut of a target at $z^{\rm ob}$ (the parabolic kernel of the production pipeline; never approximated in this note). Any galaxy that satisfies both cuts is counted, whether or not it belongs to the cluster.

Background: a scale-dependent selection bias. Richness selection is not mass selection. At fixed mass, systems whose line of sight carries excess correlated structure scatter *up* in observed richness, so a sample selected at $\lambda^{\rm ob}$ is preferentially drawn from overdense environments and clusters more strongly than a mass-matched sample. The effect is scale-dependent. At large separations the selected sample behaves as a linear tracer with a boosted, constant bias amplitude. At small separations — within a few richness apertures — the very structure that inflated $\lambda^{\rm ob}$ sits inside or near the measurement itself, and the effective amplitude differs again. We encode this in a redMaPPer selection bias $b_{\rm rm}(\theta, \lambda^{\rm ob}, z^{\rm ob})$, defined as the effective tracer bias of the selected sample in its cross-correlation with halos of mass M:

$$\xi_{\rm rm,h}(\theta; M, z) = b(M, z)\, b_{\rm rm}(\theta, \lambda^{\rm ob}, z^{\rm ob})\, \xi_{\rm mm}\big(|\boldsymbol{r}_{\rm rm} - \boldsymbol{r}_{\rm h}|\big), \tag{1}$$

with $b(M, z)$ the halo bias of the correlated structure, $\xi_{\rm mm}$ the nonlinear matter–matter correlation function (halofit), and $|\boldsymbol{r}_{\rm rm} - \boldsymbol{r}_{\rm h}|$ the 3-D comoving separation between the target and the halo — both tracers are biased, and the cross-correlation carries both biases. Following the simulation-calibrated model of Costanzi et al., we take $b_{\rm rm}$ to be two plateaus bridged by a fixed sigmoid,

$$b_{\rm rm}(\theta, \lambda^{\rm ob}, z^{\rm ob}) = b^{\rm ss}_{\rm rm}\,[1 - \sigma(\theta)] + b^{\rm ls}_{\rm rm}\,\sigma(\theta), \qquad \sigma(\theta) = \Big[1 + e^{-k(\theta-\theta_0)}\Big]^{-1}, \tag{2}$$

with $k = 2.5/\theta_{\lambda,\rm ob}$ and $\theta_0 = \theta_{\lambda,\rm ob}/2$, where $\theta_{\lambda,\rm ob}$ is the target's aperture angle: σ runs from $\simeq 0$ well inside the aperture to 1 at large separations, so $[1 - \sigma]$ tags the small-scale regime governed by the plateau $b^{\rm ss}_{\rm rm}$ and σ the large-scale regime governed by $b^{\rm ls}_{\rm rm}$. The transition at half the aperture, and the sharpness k, are held fixed; the two plateaus are the physical unknowns.

This bias is not bookkeeping: it propagates directly into the stacked weak-lensing observable. The selected sample's large-scale-structure lensing contribution — the revisited second-halo term, under the correlation model of eq. (1) — reads

$$\begin{aligned}\langle \Delta\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle = 2\pi \int d\theta\, \sin\theta \int dz\, \frac{dV}{dz\, d\Omega} \int dM\, n(M, z)& \\ \times\ \big[\, 1 + b(M, z)\, b_{\rm rm}(\theta, \lambda^{\rm ob}, z^{\rm ob})\, \xi_{\rm mm}\big(|\boldsymbol{r}_{\rm rm} - \boldsymbol{r}_{\rm h}|\big)\, \big]& \\ \times\ \Delta\Sigma_{\rm mis}\big(R \mid M, z, R_{\rm mis} = \theta\, \chi(z^{\rm ob})\big),& \end{aligned} \tag{3}$$

where $\Delta\Sigma_{\rm mis}$ is the excess surface density of a halo of mass M miscentred by the transverse offset $R_{\rm mis}$ corresponding to the angular separation θ, and the integral is understood to be normalised by the same (z, M) halo measure. Every projected structure that lenses the source galaxies behind the target is weighted by exactly the $b_{\rm rm}(\theta)$ defined above: an error in the selection-bias plateaus is an error in the cluster lensing profile, and hence in the mass calibration.

The plateaus are not free functions — the remainder of this section shows they are fixed by requiring the projected-galaxy budget of the sample to close.

Deriving the scale-dependent bias. The observed richness can be split as

$$\lambda^{\rm ob} \ = \ \lambda^{\rm tr} \ + \ \langle \Delta^{\rm prj} \rangle, \tag{4}$$

with $\Delta^{\rm prj}$ the number of *projected* galaxies: members of other halos in the target's photo-z cylinder whose sky positions fall inside the target's aperture. Specifically, for clusters at redshift $z^{\rm ob}$ with observed richness $\lambda^{\rm ob}$ and true richness $\lambda^{\rm tr}$, the mean contamination due to projected systems reads

$$\langle \Delta^{\rm prj} \rangle(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}) = 2\pi \int_0^{\pi} d\theta \sin\theta\ \langle \Delta\lambda \rangle_\theta\,, \tag{5}$$

where $\langle \Delta\lambda \rangle_\theta$ is the mean projected richness per unit solid angle contributed at angular separation θ from the target centre. It encodes the redMaPPer–halo association over the photo-z cylinder:

$$\langle \Delta\lambda \rangle_\theta \ \equiv\ \langle \Delta\lambda \rangle_{\rm rnd} \ +\ b^{\rm ss}_{\rm rm}\,[1 - \sigma(\theta)]\ \langle \widetilde{\Delta\lambda}(\theta) \rangle_{\rm lss} \ +\ b^{\rm ls}_{\rm rm}\,\sigma(\theta)\ \langle \widetilde{\Delta\lambda}(\theta) \rangle_{\rm lss} \tag{6}$$

$$\begin{aligned} &= \int dz\, \frac{dV}{dz\, d\Omega}(z)\, w(z, z^{\rm ob}) \int dM\, n(M, z)\, \big[1 + \xi_{\rm rm,h}(\theta; M, z)\big] \\ &\qquad \times \int_0^{\lambda^{\rm ob}} d\lambda\, P(\lambda \mid M, z)\, \lambda\, f_A\big(\theta, \theta_{\lambda,\rm ob}, \theta_\lambda(\lambda, z)\big), \end{aligned} \tag{7}$$

reading right to left: each halo of mass M at redshift z hosts λ galaxies with probability $P(\lambda \mid M, z)$; a fraction f_A of them falls inside the target aperture; halos are counted by the mass function $n(M, z)$ and placed either uniformly along the cylinder (the 1) or clustered around the target (the $\xi_{\rm rm,h}$); the photo-z window w selects in redshift. Two definitions:

- **Eligibility** ($\lambda < \lambda^{\rm ob}$)**.** REDMAPPER processes halos in decreasing richness order and lets richer systems claim member galaxies first, so only halos with $\lambda < \lambda^{\rm ob}$ still have galaxies left to contaminate the target — the upper limit of every λ-integral is the percolation cut, not a convenience.

- **Capture fraction** f_A**.** A projected halo's λ galaxies are spread uniformly over its own disk of angular radius $\theta_\lambda(\lambda, z)$; the target aperture (radius $\theta_{\lambda,\rm ob}$) captures the fraction $f_A = A_{\rm ov}/(\pi\theta_\lambda^2) \in [0, 1]$, with $A_{\rm ov}$ the two-disk overlap area at centre separation θ. f_A vanishes for $\theta > \theta_{\lambda,\rm ob} + \theta_\lambda$, so the support of eq. (5) ends at $\theta \le 2\theta_{\lambda,\rm ob}$ despite the formal π upper limit. Because θ_λ depends on the projected halo's own λ, f_A sits inside the λ-integral.

Substituting the correlation model eqs. (1)–(2) into eq. (7) produces exactly the decomposition announced in eq. (6), with the two channel profiles defined by the weight they carry under the same (z, M, λ)-integral:

$$\langle\Delta\lambda\rangle_{\rm rnd} = \int dz\,\frac{dV}{dz\,d\Omega}\,w\int dM\,n\int_0^{\lambda^{\rm ob}} d\lambda\,P\,\lambda \qquad \text{(weight 1: uniform interlopers)}, \tag{8}$$

$$\langle\widetilde{\Delta\lambda}(\theta)\rangle_{\rm lss} = \int dz\,\frac{dV}{dz\,d\Omega}\,w\int dM\,n\,b\,\xi_{\rm mm}\int_0^{\lambda^{\rm ob}} d\lambda\,P\,\lambda\,f_A \quad \text{(weight } b\,\xi_{\rm mm}\text{: clustered structures)}, \tag{9}$$

where the tilde marks that the target's own bias $b_{\rm rm}$ has been divided out ($b_{\rm rm}$ set to one): $\langle\widetilde{\Delta\lambda}(\theta)\rangle_{\rm lss}$ is a pure integral, computable without knowing the selection bias, and the unknown plateaus $b^{\rm ss}_{\rm rm}$, $b^{\rm ls}_{\rm rm}$ multiply it from outside. Note that the uniform channel carries no capture fraction and no θ: smearing a uniform field over the donors' disks leaves it uniform, and f_A integrates out exactly — $\int d^2\theta\,f_A = \pi\theta^2_{\lambda,\rm ob}$ for every donor size (the total-donation identity) — so $\langle\Delta\lambda\rangle_{\rm rnd}$ is the mean eligible richness per unit solid angle in the cylinder, constant in θ, and comes out of the sky integral of eq. (5).

Assembled contamination. Inserting eq. (6) into eq. (5) and integrating over the sky:

$$\langle\Delta^{\rm prj}\rangle \;=\; \langle\Delta^{\rm prj}\rangle_{\rm rnd} \;+\; \langle\Delta^{\rm prj}\rangle^{\rm ss}_{\rm lss} \;+\; \langle\Delta^{\rm prj}\rangle^{\rm ls}_{\rm lss}, \tag{10}$$

with the three members

$$\langle\Delta^{\rm prj}\rangle_{\rm rnd} = \pi\,\theta^2_{\lambda,\rm ob}\,\langle\Delta\lambda\rangle_{\rm rnd}, \tag{11}$$

$$\langle\Delta^{\rm prj}\rangle^{\rm ss}_{\rm lss} = b^{\rm ss}_{\rm rm}\,\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss}_{\rm lss}, \qquad \langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss}_{\rm lss} = 2\pi\int_0^{\pi} d\theta\,\sin\theta\,[1-\sigma(\theta)]\,\langle\widetilde{\Delta\lambda}(\theta)\rangle_{\rm lss}, \tag{12}$$

$$\langle\Delta^{\rm prj}\rangle^{\rm ls}_{\rm lss} = b^{\rm ls}_{\rm rm}\,\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ls}_{\rm lss}, \qquad \langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ls}_{\rm lss} = 2\pi\int_0^{\pi} d\theta\,\sin\theta\,\sigma(\theta)\,\langle\widetilde{\Delta\lambda}(\theta)\rangle_{\rm lss}. \tag{13}$$

$\langle\Delta^{\rm prj}\rangle_{\rm rnd}$ counts the uniform (random) interlopers — the constant $\langle\Delta\lambda\rangle_{\rm rnd}$ times the aperture area, per the total-donation identity above; $\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss}_{\rm lss}$ and $\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ls}_{\rm lss}$ are the small- and large-scale halves of the clustered (large-scale-structure) contamination, computed numerically with unit selection bias. In the production code these are `P1`, $I_2 - I_1$ and I_1 respectively.

Closing the selection bias. The plateaus follow from two conditions. First, the large-scale plateau is anchored to the halo bias of a mass-selected sample at the same $(\lambda^{\rm ob}, z^{\rm ob})$,

$$b_{\rm halo}(\lambda^{\rm ob}, z^{\rm ob}) = \frac{\int d\ln M\;M\,n(M, z^{\rm ob})\,b(M, z^{\rm ob})\,P(\lambda^{\rm ob}\mid M, z^{\rm ob})}{\int d\ln M\;M\,n(M, z^{\rm ob})\,P(\lambda^{\rm ob}\mid M, z^{\rm ob})}, \tag{14}$$

through the Buzzard-calibrated empirical relation

$$\begin{aligned} b_{\rm rm}^{\rm ls} &= b_{\rm halo}\,[1 + 0.13\,\delta_{\rm prj}]\,, \qquad \delta_{\rm prj} = \frac{\lambda^{\rm ob} - \lambda^{\rm tr}}{\langle\Delta^{\rm prj}\rangle_{\rm halo}} - 1, \\ \langle\Delta^{\rm prj}\rangle_{\rm halo} &= \langle\Delta^{\rm prj}\rangle_{\rm rnd} + b_{\rm halo}\big[\langle\widetilde{\Delta}^{\rm prj}\rangle_{\rm lss}^{\rm ss} + \langle\widetilde{\Delta}^{\rm prj}\rangle_{\rm lss}^{\rm ls}\big], \end{aligned} \tag{15}$$

where $\langle\Delta^{\rm prj}\rangle_{\rm halo}$ is the contamination a *mass-selected* target would suffer — eq. (10) with the constant $b_{\rm halo}$ in place of both plateaus — and $\delta_{\rm prj}$ measures the fractional excess of the observed inflation over it (the calibration of the 0.13 slope is tied to this denominator). Second, the observed inflation must be exactly accounted for: substituting eqs. (11)–(13) into eqs. (4) and (10),

$$\lambda^{\rm ob} - \lambda^{\rm tr} = \langle\Delta^{\rm prj}\rangle_{\rm rnd} + b_{\rm rm}^{\rm ls}\,\langle\widetilde{\Delta}^{\rm prj}\rangle_{\rm lss}^{\rm ls} + b_{\rm rm}^{\rm ss}\,\langle\widetilde{\Delta}^{\rm prj}\rangle_{\rm lss}^{\rm ss}, \tag{16}$$

which closes the small-scale plateau as the remaining linear unknown:

$$\boxed{b_{\rm rm}^{\rm ss} \;=\; \frac{(\lambda^{\rm ob} - \lambda^{\rm tr}) - \langle\Delta^{\rm prj}\rangle_{\rm rnd} - \langle\Delta^{\rm prj}\rangle_{\rm lss}^{\rm ls}}{\langle\widetilde{\Delta}^{\rm prj}\rangle_{\rm lss}^{\rm ss}}.} \tag{17}$$

Algorithm. The computation at fixed $(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})$ is therefore four numerical steps and one closure:

1. compute $b_{\rm halo}(\lambda^{\rm ob}, z^{\rm ob})$, eq. (14) — a 1-D mass integral;
2. compute $\langle\Delta^{\rm prj}\rangle_{\rm rnd}$, eq. (11) — the constant $\langle\Delta\lambda\rangle_{\rm rnd}$ times the aperture area;
3. compute the unit-bias clustered integrals $\langle\widetilde{\Delta}^{\rm prj}\rangle_{\rm lss}^{\rm ss}$ and $\langle\widetilde{\Delta}^{\rm prj}\rangle_{\rm lss}^{\rm ls}$, eqs. (12)–(13);
4. form $\langle\Delta^{\rm prj}\rangle_{\rm halo}$, $\delta_{\rm prj}$ and the large-scale plateau $b_{\rm rm}^{\rm ls}$, eq. (15);
5. close the small-scale plateau $b_{\rm rm}^{\rm ss}$, eq. (17).

With $b_{\rm rm}^{\rm ss}$ and $b_{\rm rm}^{\rm ls}$ in hand, eq. (2) returns the full scale-dependent selection bias $b_{\rm rm}(\theta, \lambda^{\rm ob}, z^{\rm ob})$.

2 The frozen-physics algorithm

The operators of §1 are 4-D quadratures over (z, M, λ, θ), and in the original variables the integrand is numerically hostile: the halo-exclusion condition, sliced along z, produces two sharp peaks bridged by a plateau in the z-integrand, on a scale $\delta z \sim 10^{-3}$. This section derives the equivalent — but numerically friendly — forms actually implemented, built on one physical observation and one change of variables.

2.1 Freezing the slow redshift dependences

The redshift barely varies across the scales that matter. The redMaPPer photo-z window, $\sigma_z(z)$ of `photoz.py` (the DES-Y3 richness-estimator kernel, read off a calibration table — not a single-galaxy photo-z error), has width $\sigma_z \simeq 0.09$ at $z^{\rm ob} = 0.5$, i.e. comoving $\sigma_\chi \simeq 205\,h^{-1}\,$Mpc ($\sigma_\chi \equiv \frac{1}{2}[\chi(z^{\rm ob} + \sigma_z) - \chi(z^{\rm ob} - \sigma_z)]$). It is much wider than the naive single-galaxy estimate $\sigma_z^{\rm gal} \sim 0.01(1+z) \simeq 0.013$ at $z = 0.3$ ($\sigma_\chi^{\rm gal} \simeq 34\,h^{-1}\,$Mpc): $\sigma_z(z)$ is the scatter of the richness *estimator* itself, built from many member galaxies and their individual photo-z's, not the redshift error of any one galaxy. Meanwhile the exclusion structure lives on $R_{\rm excl} = R_\lambda(\lambda^{\rm ob})(1+z^{\rm ob}) \simeq 1.1\,h^{-1}\,$Mpc ($\delta z \simeq 5\times10^{-4}$); the astrophysical weights — $n(M,z)$, $b(M,z)$, $P(\lambda \mid M, z)$, the aperture $\theta_\lambda(\lambda, z)$ — evolve on $\Delta z \sim 0.1$–1. We therefore *freeze* the slow weights at $z^{\rm ob}$ inside the exclusion-affected zone ($|\chi(z)-\chi_o| \lesssim 20R_{\rm excl}$, i.e. $\Delta z \lesssim 10^{-2}$; $\chi_o \equiv \chi(z^{\rm ob})$). Two protections control the error: the drift itself is $\partial_z \ln(\cdot)\,\Delta z \lesssim 10^{-2}$, and its leading term is *odd* in $\chi(z)-\chi_o$ while the correlation kernel is even, so the linear error integrates to zero and only the quadratic survives, $\sim 10^{-4}$. Direct validation against `scipy.quad` puts the freeze at 0.001–0.003% on the clustered operators across the DES-Y3 range. The photo-z window $w(z, z^{\rm ob})$ is *never* frozen — it is the one genuine line-of-sight structure and stays explicit in every integral below.

2.2 Sky variables

Both sharp structures — the exclusion ball and the correlation function — depend only on the 3-D separation between target and halo, while the aperture is transverse and the photo-z window is line-of-sight. Decompose the separation accordingly: a line-of-sight component and an exact sky-plane (transverse) component,

$$\pi \equiv \chi(z) - \chi_o, \qquad s \equiv 2\sqrt{\chi\chi_o}\,\sin(\theta/2) \quad \Longrightarrow \quad r^2 = \pi^2 + s^2 \;\;\text{(exact)}, \tag{18}$$

where $r = |\boldsymbol{r}_{\rm rm} - \boldsymbol{r}_{\rm h}|$ is the comoving separation entering $\xi_{\rm mm}$. The measure maps exactly as well:

$$dz\,\frac{dV}{dz\,d\Omega}\,2\pi \sin\theta\,d\theta = 2\pi\Big(1+\tfrac{\pi}{\chi_o}\Big)\,d\pi\;s\,ds \tag{19}$$

(the 2π prefactors are the number; the line-of-sight variable π always carries a differential or an argument). In these variables the geometry is transparent. The aperture support $f_A = 0$ beyond $\theta_\lambda + \theta_{\lambda,\rm ob} \leq 2\theta_{\lambda,\rm ob}$ becomes the cylinder $s \leq 2R_{\rm excl}$; the halo-exclusion condition ($\xi_{\rm mm} \equiv 0$ for separations below $R_{\rm excl}$, §1) becomes the clean spherical statement $r > R_{\rm excl}$ — a hard *lower limit* of the radial integral, not a discontinuity inside the domain. The twin peaks of the original z-integrand were never physical structure: they are the shadow of this excluded ball sliced along z, and they never form in (π, s).

2.3 Reduced operators

At each $(\lambda^{\rm ob}, z^{\rm ob})$ the algorithm evaluates the random channel and the clustered channel, the latter split into two line-of-sight zones:

$$\langle\Delta^{\rm prj}\rangle_{\rm rnd} = \pi\,\theta^2_{\lambda,\rm ob}\int dz\,\frac{dV}{dz\,d\Omega}\,w(z,z^{\rm ob})\,\langle\Delta\lambda\rangle(z), \qquad \langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss}_{\rm lss} = \langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss,excl}_{\rm lss} + \langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss,cyl}_{\rm lss}. \tag{20}$$

(The fiducial pipeline additionally removes the halo-exclusion sphere from the random channel; in the sky variables this carve-out is the incomplete transverse moment $M_0(2) - M_0(x_{\rm excl}(\pi))$, with $M_0(y) = \int_0^y x\,I_0(x)\,dx$ over the unweighted analogue of eq. 21 and $x_{\rm excl} = \sqrt{\max(0, 1-(\pi/R_{\rm excl})^2)}$ — a ring-sliver correction of a few $\times 10^{-3}$ relative, adopted by the implementation so that all three operators share one exclusion convention.) The *exclusion zone*, $|\pi| \leq \pi_s \equiv 20\,R_{\rm excl}$, is the cluster's own neighbourhood: it contains the halo-exclusion sphere $r < R_{\rm excl}$ and the target aperture, and every calculation there must track the boundary $r = R_{\rm excl}$. The *free-of-exclusion zone* is the rest of the photo-z cylinder, out to the edges of the window: beyond π_s, $r \geq |\pi| > \pi_s = 20R_{\rm excl} \gg R_{\rm excl}$ automatically, so the exclusion condition is satisfied by every point and never has to be imposed — there is no boundary left to track. Both zones stay inside the transverse cylinder $s \leq 2R_{\rm excl}$; only the exclusion *sphere* is absent from the second. Throughout we write the small-scale (ss) operator; the large-scale one is the same expression with $I^{\rm ss}_\lambda \to I^{\rm ls}_\lambda$.

The split point balances two competing requirements. The exclusion zone must stay narrow enough in redshift that the astrophysical weights can be frozen at $z^{\rm ob}$ ($\Delta z \lesssim 10^{-2}$, §2.1); the free-of-exclusion zone must start far enough out that $s \ll |\pi|$ and the transverse expansion of eq. (26) converges. At $\pi_s = 20R_{\rm excl}$ both are simultaneously controlled: the largest ratio in the free-of-exclusion zone is $(2R_{\rm excl}/\pi_s)^2 = 10^{-2}$, the $B_{\rm ss}$ term of eq. (29) retains the first correction at exactly this order, and the next omitted contribution is $(2R_{\rm excl}/\pi_s)^4 \lesssim 10^{-4}$. At the reference $R_{\rm excl} \simeq 1.1\,h^{-1}$ Mpc the geometry is $s_{\rm max} \simeq 2.2\,h^{-1}$ Mpc, $\pi_s \simeq 22\,h^{-1}$ Mpc, $r_{\rm max} = \sqrt{22^2 + 2.2^2} \simeq 22.1\,h^{-1}$ Mpc: the exclusion zone is a short central segment of a much longer photo-z cylinder ($\sigma_\chi \simeq 205\,h^{-1}$ Mpc per side).

The building block of both regimes is the frozen mass-integral moment (Fig. 1),

$$\boxed{I_\lambda(x) = \int dM\, n(M, z^{\rm ob})\, b(M, z^{\rm ob}) \int_0^{\lambda^{\rm ob}} d\lambda\, P(\lambda \mid M, z^{\rm ob})\, \lambda\, f_A(x, x_\lambda),} \qquad x = \tfrac{s}{R_{\rm excl}}, \quad x_\lambda = \left(\tfrac{\lambda}{\lambda^{\rm ob}}\right)^{0.2}, \tag{21}$$

with the channel versions carrying the sigmoid weight,

$$I_\lambda^{\rm ss}(x) = [1 - \sigma(x)]\, I_\lambda(x), \qquad I_\lambda^{\rm ls}(x) = \sigma(x)\, I_\lambda(x). \tag{22}$$

Two contamination densities accompany it: the unweighted one that drives the random channel, and its bias-weighted counterpart that sets the amplitude in regime B,

$$\begin{aligned} \langle \Delta\lambda \rangle(z) &= \int dM\, n(M, z) \int_0^{\lambda^{\rm ob}} d\lambda\, P(\lambda \mid M, z)\, \lambda, \\ \langle \Delta\lambda \rangle_b(z) &= \int dM\, n(M, z)\, b(M, z) \int_0^{\lambda^{\rm ob}} d\lambda\, P(\lambda \mid M, z)\, \lambda. \end{aligned} \tag{23}$$

These are not new objects. Eligibility makes every donor disk smaller than the target aperture ($x_\lambda \le 1$), so $f_A(0, x_\lambda) = 1$: at zero transverse offset the mass-integral moment captures every donated galaxy, and

$$I_\lambda(0) = \langle \Delta\lambda \rangle_b(z^{\rm ob}) \tag{24}$$

— the densities are the $\theta = 0$, $z = z^{\rm ob}$ limits of the frozen profiles ($\langle \Delta\lambda \rangle(z^{\rm ob})$ likewise from the unweighted analogue of eq. 21).

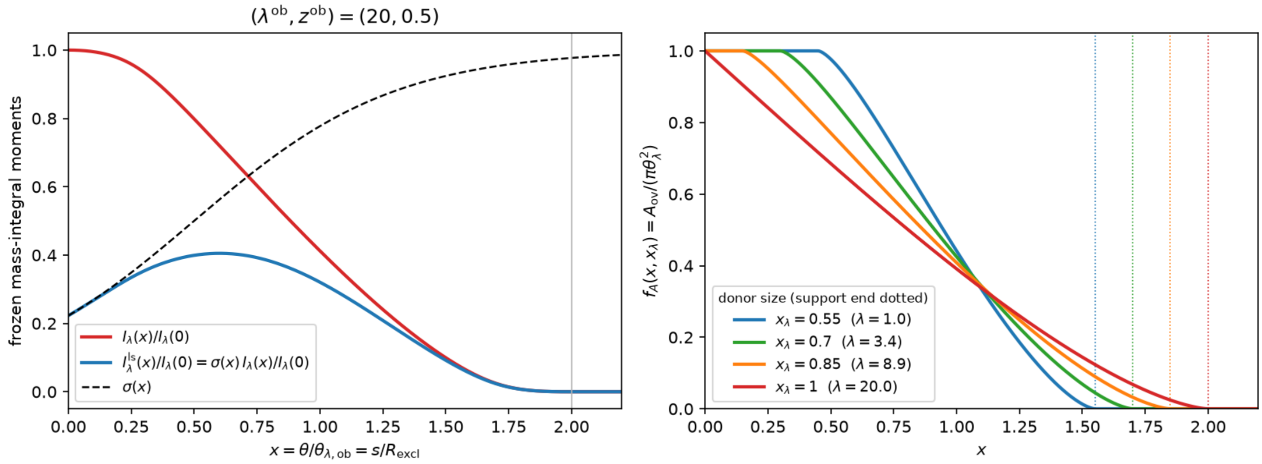

Figure 1: *Left:* the frozen mass-integral moment at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$: $I_\lambda(x)$ (red, normalised to $x = 0$), its large-scale part $I_\lambda^{\rm ls} = \sigma I_\lambda$ (blue), and the sigmoid $\sigma(x)$ (dashed); $I_\lambda^{\rm ss}$ is the difference of the two curves. Support ends at $x = 1 + x_{\lambda,\rm max} \le 2$. *Right:* the capture fraction $f_A(x, x_\lambda)$ of §1 for several donor sizes; I_λ is their bias- and richness-weighted average.

Exclusion zone. The halo-exclusion sphere is spherical in (π, s), so integrate on the sphere: with $\mu = \pi/r$,

$$\boxed{\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss,excl}_{\rm lss} = 2\pi\int_{R_{\rm excl}}^{r_{\rm max}} dr\, r^2\, \xi_{\rm mm}(r)\, G_{\rm ss}(r), \qquad G_{\rm ss}(r) = 2\int_{\mu_{\rm lo}(r)}^{\mu_{\rm hi}(r)} d\mu\, I^{\rm ss}_\lambda\Big(\tfrac{r}{R_{\rm excl}}\sqrt{1-\mu^2}\Big)\, \bar{w}(r\mu),} \tag{25}$$

where $\bar{w}$ is the two-sided average of the photo-z window and the μ-limits trace the domain boundaries: the cylinder wall gives $\mu_{\rm lo}(r) = \sqrt{\max(0,\, 1-(2R_{\rm excl}/r)^2)}$, the zone split gives $\mu_{\rm hi}(r) = \min(1,\, \pi_s/r)$, and the radial integral ends at $r_{\rm max} = \sqrt{\pi_s^2 + 4R_{\rm excl}^2}$, the corner of the domain. The integrand is smooth (Fig. 2): the exclusion is the hard lower limit $r = R_{\rm excl}$, and the only non-analytic point is a derivative kink at $r = 2R_{\rm excl}$ where the cylinder wall engages — split the grid there and log-spaced Gauss–Legendre nodes converge everywhere.

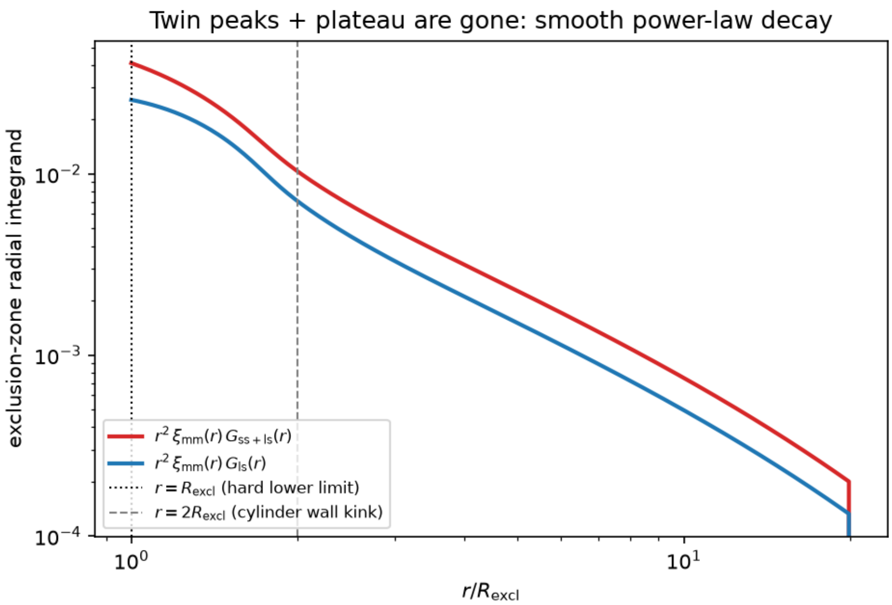

Figure 2: Exclusion-zone radial integrand $r^2\,\xi_{\rm mm}(r)\,G(r)$ of eq. (25) at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$, for the total (ss+ls) and large-scale mass integrals. Exclusion radius = hard lower limit (dotted); derivative kink at $r = 2R_{\rm excl}$ (dashed). The twin peaks and ring plateau of the original z-integrand are absent.

Free-of-exclusion zone. Beyond π_s the geometry is Limber-like: the cylinder is *thin*. Its radius, $2R_{\rm excl} \simeq 2\,h^{-1}\,$Mpc, is small against every scale that varies along the line of sight ($|\pi| \geq \pi_s \simeq 22\,h^{-1}\,$Mpc, photo-$z$ window $\sigma_\chi \simeq 205\,h^{-1}\,$Mpc), so expand the separation about the cylinder axis,

$$r = |\pi|\sqrt{1+s^2/\pi^2} = |\pi| + \frac{s^2}{2|\pi|} + O\big(s^4/|\pi|^3\big), \qquad \xi_{\rm mm}(r) = \xi_{\rm mm}(|\pi|) + \frac{s^2}{2|\pi|}\,\xi'_{\rm mm}(|\pi|) + O\Big[\big(\tfrac{2R_{\rm excl}}{\pi}\big)^4\Big]. \tag{26}$$

This is the Limber step: the correlation across the thin cross-section is replaced by its value and slope on the axis. With the exclusion sphere gone (no $r = R_{\rm excl}$ boundary to track here), the s-integral of eq. (31) acts directly on this expansion and reduces to two fixed moments of the mass integral,

$$\int_0^{2R_{\rm excl}} s\,ds\, I^{\rm ss}_\lambda(x)\,\xi_{\rm mm}(r) = A_{\rm ss}\,\xi_{\rm mm}(|\pi|) + \frac{B_{\rm ss}}{2|\pi|}\,\xi'_{\rm mm}(|\pi|) + O\Big[\big(\tfrac{2R_{\rm excl}}{\pi}\big)^4\Big],$$
$$A_{\rm ss} = R_{\rm excl}^2\int_0^2 x\, I^{\rm ss}_\lambda(x)\,dx, \qquad B_{\rm ss} = R_{\rm excl}^4\int_0^2 x^3\, I^{\rm ss}_\lambda(x)\,dx. \tag{27}$$

Two z-dependences remain to be restored before this can be integrated down the line of sight. Geometric: the aperture is fixed in *angle*, so the transverse variable dilates, $x = (s/R_{\rm excl})\sqrt{\chi_o/\chi}$, and each moment above picks up a power of χ/χ_o per two powers of s — combined with the $(1 + \pi/\chi_o)$ of the measure eq. (19), this is the squared distance ratio below. Astrophysical: the weights are no longer frozen, but their *shape* drifts far more slowly than their *amplitude*, so keep the amplitude exactly and freeze only the shape,

$$I_\lambda(x; z) \;\simeq\; \frac{\langle\Delta\lambda\rangle_b(z)}{\langle\Delta\lambda\rangle_b(z^{\rm ob})}\, I_\lambda(x; z^{\rm ob}), \tag{28}$$

anchored by eq. (24) ($I_\lambda(0; z) = \langle\Delta\lambda\rangle_b(z)$ exactly). Writing the background side ($z > z^{\rm ob}$, at $z_+ = z(\chi_o + \pi)$) and the foreground side ($z < z^{\rm ob}$, at $z_- = z(\chi_o - \pi)$) explicitly:

$$\begin{aligned}\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss,cyl}_{\rm lss} &= 2\pi \int_{\pi_s}^{\chi(z_{\rm hi})-\chi_o} d\pi \left(1 + \tfrac{\pi}{\chi_o}\right)^2 w(z_+, z^{\rm ob})\, \frac{\langle\Delta\lambda\rangle_b(z_+)}{\langle\Delta\lambda\rangle_b(z^{\rm ob})} \left[A_{\rm ss}\,\xi_{\rm mm}(\pi) + \frac{B_{\rm ss}}{2\pi}\,\xi'_{\rm mm}(\pi)\right] \\ &+ 2\pi \int_{\pi_s}^{\chi_o-\chi(z_{\rm lo})} d\pi \left(1 - \tfrac{\pi}{\chi_o}\right)^2 w(z_-, z^{\rm ob})\, \frac{\langle\Delta\lambda\rangle_b(z_-)}{\langle\Delta\lambda\rangle_b(z^{\rm ob})} \left[A_{\rm ss}\,\xi_{\rm mm}(\pi) + \frac{B_{\rm ss}}{2\pi}\,\xi'_{\rm mm}(\pi)\right], \end{aligned} \tag{29}$$

where $z_{\rm lo}$ and $z_{\rm hi}$ are the edges of the photo-z window (where w vanishes). The zone split at $\pi_s = 20R_{\rm excl}$ makes the expansion parameter of eq. (26) equal to $(2R_{\rm excl}/\pi_s)^2 = 10^{-2}$ at the split, decreasing outward; with the first moment correction retained, the residual is $O(10^{-4})$. (The channel sum of the leading moments is closed analytically by the total-donation identity, $A_{\rm ss} + A_{\rm ls} = R^2_{\rm excl}\,\langle\Delta\lambda\rangle_b(z^{\rm ob})/2$ — a free consistency check on the numerics.) The free-of-exclusion zone cannot be truncated early (Fig. 3): the BAO feature sits well inside the photo-z window.

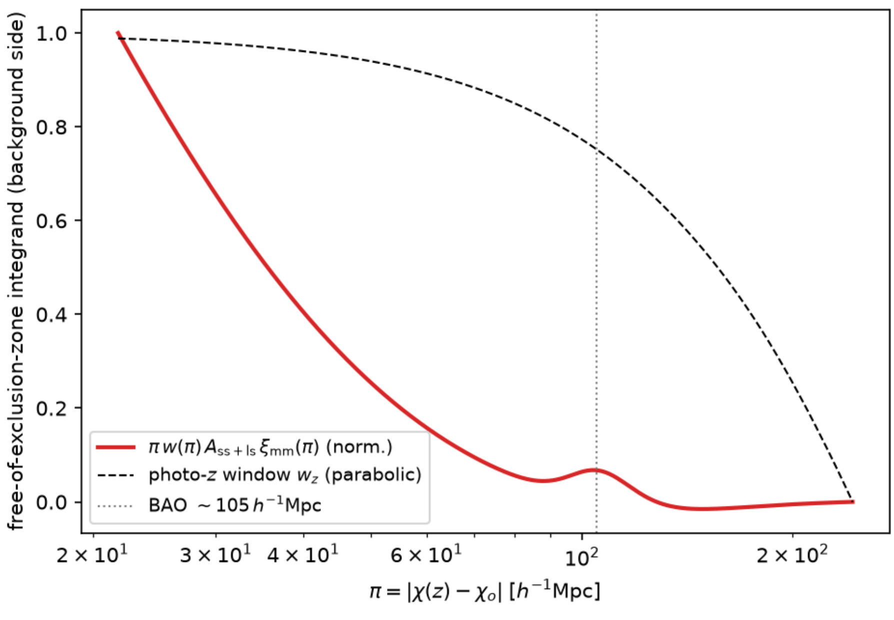

Figure 3: Free-of-exclusion-zone line-of-sight integrand of eq. (29) (background side, log-π measure). The BAO feature (dotted) lies inside the photo-z window (dashed): the far tail carries real weight, hence both integrals run to the edges of the window.

Derivation. Combining eqs. (9) and (12):

$$\begin{aligned}\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss}_{\rm lss} = 2\pi \int_0^{\pi} d\theta \sin\theta\, [1 - \sigma(\theta)] \int dz\, \frac{dV}{dz\, d\Omega}\, w(z, z^{\rm ob}) \int dM\, n(M, z)\, b(M, z)\, \xi_{\rm mm}(r) \\ \times \int_0^{\lambda^{\rm ob}} d\lambda\, P(\lambda \mid M, z)\, \lambda\, f_A. \end{aligned} \tag{30}$$

The sky variables eqs. (18)–(19) are exact and give

$$\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ss}_{\rm lss} = 2\pi\int d\pi\,\left(1+\tfrac{\pi}{\chi_o}\right)w\big(z(\pi),z^{\rm ob}\big)\int_0^{2R_{\rm excl}} s\,ds\,[1-\sigma]\,\xi_{\rm mm}\big(\sqrt{\pi^2+s^2}\big)\int dM\,n\,b\int_0^{\lambda^{\rm ob}} d\lambda\,P\,\lambda\,f_A. \tag{31}$$

Freeze. For $|\pi|\le\pi_s$ the slow weights are evaluated at $z^{\rm ob}$ (§2.1). $\xi_{\rm mm}$ carries no M-dependence, so it passes through the mass integral and the (M,λ) block of eq. (31) contracts, with the sigmoid universal in x:

$$[1-\sigma(\theta)]\int dM\,n\,b\int_0^{\lambda^{\rm ob}} d\lambda\,P\,\lambda\,f_A \xrightarrow{z\to z^{\rm ob}} [1-\sigma(x)]\,I_\lambda(x) \;=\; I^{\rm ss}_\lambda(x). \tag{32}$$

Exclusion zone. The frozen integrand depends on (π,s) only through $r=\sqrt{\pi^2+s^2}$ and $x=s/R_{\rm excl}$, i.e. through r and the direction cosine $\mu=\pi/r$. The spherical map $d\pi\,s\,ds=r^2\,dr\,d\mu$ turns the domain $\{r>R_{\rm excl},\ s\le 2R_{\rm excl},\ |\pi|\le\pi_s\}$ into the limits $\mu_{\rm lo}(r)$, $\mu_{\rm hi}(r)$, $r_{\rm max}$ of eq. (25); the factor $(1+\pi/\chi_o)$ is odd about the cluster and averages to one by parity, and the two line-of-sight signs fold into $\bar{w}$. This is eq. (25).

Free-of-exclusion zone. Here $s\le 2R_{\rm excl}\ll|\pi|$: Taylor- expanding r and $\xi_{\rm mm}(r)$ about the cylinder axis (eq. 26, the Limber step) turns the s-integral of eq. (31) into the fixed moments $A_{\rm ss}$, $B_{\rm ss}$; restoring the geometric dilation of x and the amplitude-drift ansatz eq. (28) for the astrophysical weights gives eq. (29). See §2.3 for the expansion, the two z-restorations, and the error budget in full.

Random channel. No $\xi_{\rm mm}$, no exclusion: f_A integrates out over the sky (total-donation identity) and the first member of eq. (20) follows directly from eqs. (8) and (11).

The full validation — operators against `scipy.quad`, marginalised plateaus and the end-to-end $b_{\rm rm}(\theta)$ against the production fiducial — is §3.

3 Validation

Reference implementation: `FrozenSelBias` (`src/richness_selection/frozen_bsel.py`), a drop-in for the production `SelBias` — only the computation of the three operators changes; the closure, the $\lambda^{\rm tr}$-marginalisation and the lensing pipeline are shared code. Both methods use the fiducial (Costanzi) conventions — the $\langle\Delta^{\rm prj}\rangle_{\rm halo}$ denominator of eq. (15) and the random-channel exclusion carve-out of §2.3 — so every residual below is pure operator numerics. Scripts: `validations/frozen_bsel_validation.py` (operators, plateaus, budget), `docs/make_validation_plots_frozen_physics.py` (end-to-end figures).

3.1 Operators against `scipy.quad`

Adaptive quadrature in z ($\epsilon_{\rm rel}=10^{-6}$) on the original 4-D form eq. (30), matched inner grids, against the production grid code (fiducial) and the frozen algorithm:

$\lambda^{\rm ob}$	$z^{\rm ob}$	source	$\langle\Delta^{\rm prj}\rangle_{\rm rnd}$ err	$\langle\widetilde{\Delta}^{\rm prj}\rangle^{\rm ls}_{\rm lss}$ err	(ss+ls) err
20.0	0.500	fiducial	0.018%	0.004%	0.002%
		frozen	0.006%	0.037%	0.037%
52.5	0.425	fiducial	0.007%	0.020%	0.021%
		frozen	0.015%	0.030%	0.019%
130.0	0.575	fiducial	0.004%	0.026%	0.027%
		frozen	0.017%	0.003%	0.003%

Worst frozen error 0.037%, comfortably below the 0.1% pipeline tolerance and on par with the fiducial grid code.

3.2 Marginalised plateaus and budget closure

The $\lambda^{\rm tr}$-marginalised plateaus (the scalars that feed the lensing kernel through eq. 2), fiducial → frozen:

$\lambda^{\rm ob}$	$z^{\rm ob}$	$b^{\rm ss}_{\rm rm}$	$b^{\rm ls}_{\rm rm}$	max per-$\lambda^{\rm tr}$ dev
20.0	0.500	23.476 → 23.484	3.824 → 3.824	0.06%
52.5	0.425	21.056 → 21.052	5.924 → 5.924	0.03%
130.0	0.575	55.933 → 55.957	12.440 → 12.440	0.04%

The per-$\lambda^{\rm tr}$ plateau vectors deviate by at most 0.06% anywhere on the marginalisation grid. The budget identity eq. (16) closes to machine precision ($\max|\text{budget} - (\lambda^{\rm ob} - \lambda^{\rm tr})| = 1.8 \times 10^{-15}$) — exact by construction of eq. (17), verified as an implementation check.

3.3 End-to-end: $b_{\rm rm}(\theta)$ and the lensing observable

Figure 4 sweeps the fixed-$\lambda^{\rm tr}$ profiles at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$: the frozen curves are visually coincident with production, with residuals $\leq 0.044\%$ of the profile scale — worst for the sign-changing $\Delta^{\rm prj} = 0$ profile, where the $b^{\rm ss}_{\rm rm}$ closure is a near-cancellation and amplifies operator errors; the amplification stays mild because ss and ls share one pipeline (§2.3) and their errors largely cancel in eq. (17). Figure 5 propagates the comparison through the lensing pipeline into $\langle \Delta\Sigma^{\rm prj}(R) \rangle$ (eq. 3): residuals stay below $\simeq 0.05\%$ everywhere away from the $\Delta^{\rm prj} = 0$ sign crossing.

3.4 Cost

Per $(\lambda^{\rm ob}, z^{\rm ob})$, median over repeats: frozen 9.2 ms fresh vs production 6.4 ms; both $\sim 0.3\,\mu$s on cached repeat calls (all levels of the frozen computation — the (M, λ) contraction, the line-of-sight tables, the drift splines, the assembled operators — are memoised per $(\lambda^{\rm ob}, z^{\rm ob})$). Benchmark: `validations/benchmark_frozen.py`.

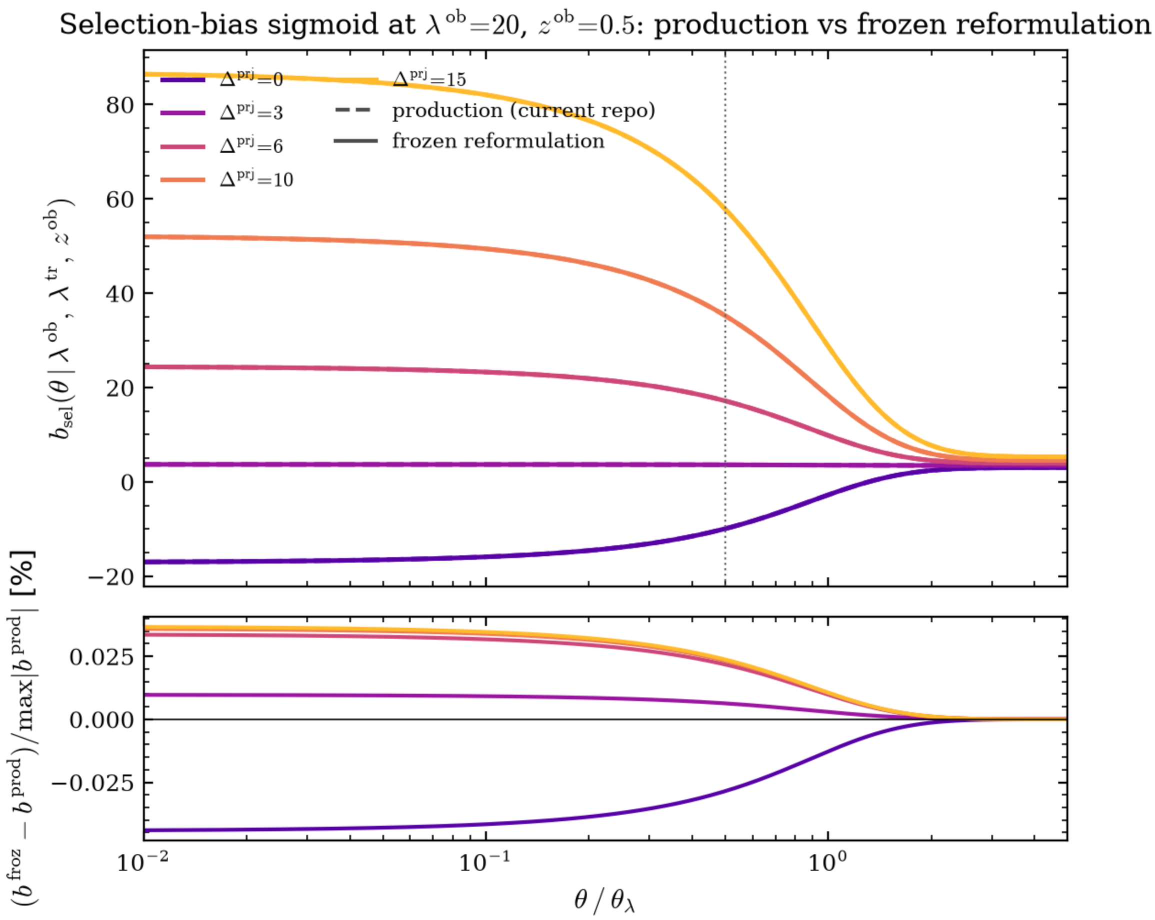

Figure 4: $b_{\rm sel}(\theta \mid \lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})$ at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$ for $\Delta^{\rm prj} \in \{0, 3, 6, 10, 15\}$: production (dashed) vs frozen (solid, visually coincident). *Bottom:* residual normalised by $\max_\theta |b^{\rm prod}|$ per $\Delta^{\rm prj}$; worst 0.044% at $\Delta^{\rm prj} = 0$.

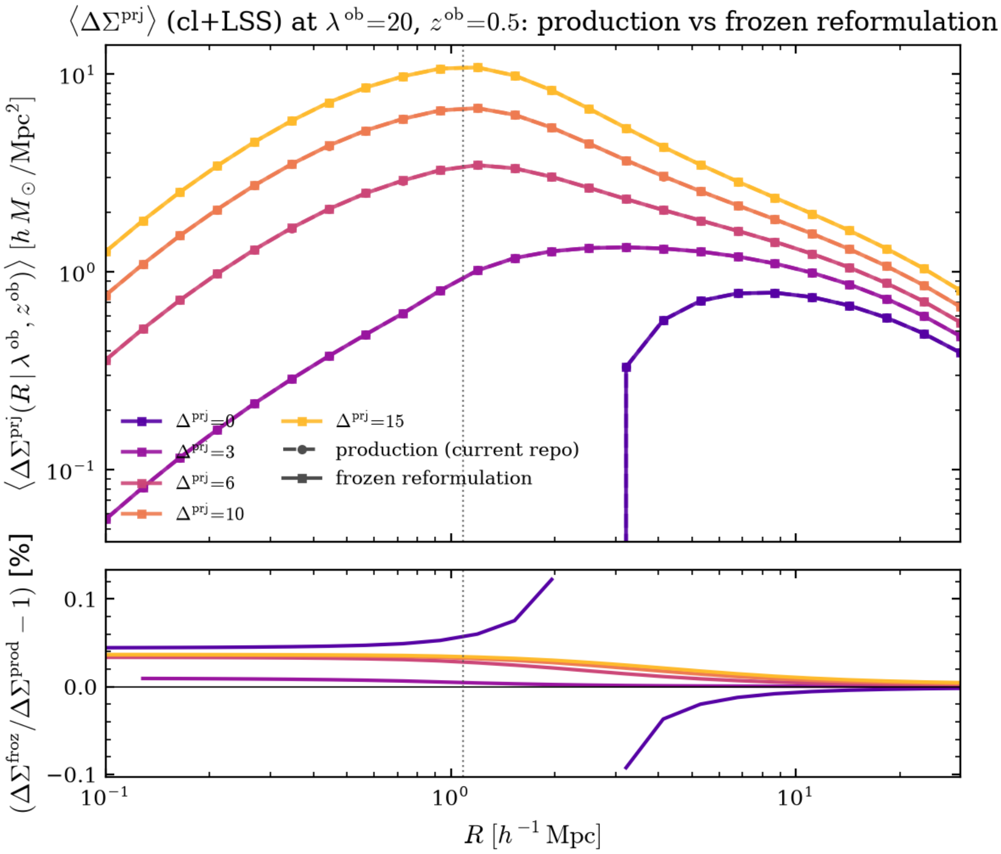

Figure 5: $\langle \Delta\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob}) \rangle$ (cl+LSS piece) at the same $(\lambda^{\rm ob}, z^{\rm ob}, \Delta^{\rm prj})$ grid, production (dashed, circles) vs frozen (solid, squares); curves overlap at plot resolution. *Bottom:* fractional residual, $\lesssim 0.05\%$ throughout. Points within 5% of the $\Delta^{\rm prj} = 0$ curve's sign crossing are dropped from the residual: the fractional ratio diverges there while the absolute difference stays at the same level as everywhere else.